

Exploratory Data Analysis

Question 1

Which precincts had the closest presidential race (between Democrats and Republicans)?

Answer 1

```
library(tidyverse)
```

```
— Attaching core tidyverse packages — tidyverse 2.0.0
—
✓ dplyr      1.1.4    ✓ readr      2.1.5
✓ forcats    1.0.1    ✓ stringr    1.5.2
✓ ggplot2    4.0.0    ✓ tibble     3.3.0
✓ lubridate  1.9.4    ✓ tidyr      1.3.1
✓ purrr      1.1.0
— Conflicts — tidyverse_conflicts()
—
* dplyr::filter() masks stats::filter()
* dplyr::lag()     masks stats::lag()
i Use the conflicted package (<http://conflicted.r-lib.org/>) to force all
conflicts to become errors
```

```
g24_sov_precinct <- read_csv("~/stat133/gerrymandering-leyna-liu/data/
g24_sov_precinct.csv")
```

```
Rows: 51123 Columns: 76
```

```
— Column specification
```

```
Delimiter: ","
```

```
chr (4): SVPREC, SVPREC_KEY, ELECTION, GEO_TYPE
```

```
dbl (72): COUNTY, FIPS, ADDIST, CDDIST, SDDIST, BEDIST, TOTREG, DEMREG,
REPR...
```

```
i Use `spec()` to retrieve the full column specification for this data.
i Specify the column types or set `show_col_types = FALSE` to quiet this
message.
```

```
# Here we use variables/columns PRSDEM01 and PRSREP01 because it shows the
votes between presidential candidates (democrats vs. republicans)
```

```
# we use abs() as the absolute value by subtracting democratic presidential
candidate votes - republican presidential candidate votes

# then we filter() to remove the precincts with no votes (or 0s)

# then we arrange() so that the closest races are first (value of 0 in the
cells first)

# and i selected specific columns so it is easier to look at

closest <- g24_sov_precinct |>
  mutate(closest_race = abs(PRSDEM01 - PRSREP01)) |>
  filter(PRSDEM01 + PRSREP01 > 0) |>
  arrange(closest_race) |>
  select(SVPREC_KEY, COUNTY, PRSDEM01, PRSREP01, closest_race)

closest
```

```
# A tibble: 33,971 × 5
  SVPREC_KEY COUNTY PRSDEM01 PRSREP01 closest_race
  <chr>      <dbl>    <dbl>    <dbl>    <dbl>
1 06001441300      1        40        40         0
2 06001482000      1        10        10         0
3 06001835310      1       148       148         0
4 06001838000      1        69        69         0
5 06001870100      1        11        11         0
6 06005CP27A       3        80        80         0
7 060071111A       4        71        71         0
8 060072201        4         7         7         0
9 060074203A       4        82        82         0
10 060075805A      4       136       136         0
# i 33,961 more rows
```

It seems like County 1 had the closest race as the votes for Democrats and Republicans were equal. Then County 3 and 4 were next in terms of race closest.

Question 2

What is the range of total sizes of the precincts?

Answer 2

```
# Here, I used the TOTVOTE column/variable because it is the total number of
votes casted per precinct (or row).

# So, by calculating the min and the max of TOTVOTE you can see the range of
votes casted per precinct or essentially the range of total sizes of
precincts.
```

```
stats_precinct <- g24_sov_precinct |>
  summarize(
    min_precinct = min(TOTVOTE, na.rm = TRUE),
    median_precinct = median(TOTVOTE, na.rm = TRUE),
    mean_precinct = mean(TOTVOTE, na.rm = TRUE),
    max_precinct = max(TOTVOTE, na.rm = TRUE)
  )

stats_precinct
```

```
# A tibble: 1 × 4
  min_precinct median_precinct mean_precinct max_precinct
      <dbl>         <dbl>         <dbl>         <dbl>
1         0             91         2202.         3793980
```

The output shows that the min is 0 and the max is 3,793,980 so that is the range. I also added the median and mean and one could also add Q1 and Q3 if they wanted to the summary statistics using the quantile function.

Question 3

How many precincts/counties/districts featured two candidates from the same party?

Answer 3

```
# So, ASSDEM01 is votes casted for first democratic candidate for assembly,
ASSDEM02 is votes casted for second democratic candidate for assembly,
CNGDEM01 is votes casted for democratic candidate for congress, CNGREP01 is
votes casted for republican candidate for congress, and so forth. It seems
like there are 2 candidates for each party running for assembly, congress, and
senate.

# First ASSDEM01 > 0 will allow R to output TRUE if anyone voted for 1st
democratic candidate running for assembly position, and FALSE if no one voted
for them. Then the same is done for ASSDEM02 (the 2nd democratic candidate).

# Then, >= 2 means if a row for example had TRUE + FALSE = 1 + 0 = 1, meaning
only ASSDEM01 got votes, then >=2 would output FALSE, meaning that only 1
candidate got votes in this party. So if the output is TRUE, that means there
are multiple candidates from the same party. The same would be repeated for
the two republican candidates (ASSREP01 and ASSREP02).

# same_assembly = dem_assembly | rep_assembly -> would output TRUE if
dem_assembly OR rep_assembly have more than 2 votes in that specific precinct
(or row), and FALSE otherwise.
```

```
# then, the exact same process is applied to the congress and senate elections.
```

```
same_party <- g24_sov_precinct |>
  mutate(
    dem_assembly = ((ASSDEM01 > 0) + (ASSDEM02 > 0)) >= 2,
    rep_assembly = ((ASSREP01 > 0) + (ASSREP02 > 0)) >= 2,
    same_assembly = dem_assembly | rep_assembly,
    dem_congress = ((CNGDEM01 > 0) + (CNGDEM02 > 0)) >= 2,
    rep_congress = ((CNGREP01 > 0) + (CNGREP02 > 0)) >= 2,
    same_congress = dem_congress | rep_congress,
    dem_senate = ((SENDEM01 > 0) + (SENDEM02 > 0)) >= 2,
    rep_senate = ((SENREP01 > 0) + (SENREP02 > 0)) >= 2,
    same_senate = dem_senate | rep_senate
  )

same_party
```

```
# A tibble: 51,123 × 85
  COUNTY FIPS SVPREC ADDIST SVPREC_KEY ELECTION GEO_TYPE CDDIST SDDIST
BEDIST
  <dbl> <dbl> <chr>    <dbl> <chr>      <chr>    <chr>    <dbl> <dbl>
<dbl>
1      1  6001 200100      14 060012001... g24      svprec      12      7
2
2      1  6001 200100A      14 060012001... g24      svprec      12      7
2
3      1  6001 200200      14 060012002... g24      svprec      12      7
2
4      1  6001 200200A      14 060012002... g24      svprec      12      7
2
5      1  6001 201400      14 060012014... g24      svprec      12      7
2
6      1  6001 201400A      14 060012014... g24      svprec      12      7
2
7      1  6001 202200      14 060012022... g24      svprec      12      7
2
8      1  6001 202200A      14 060012022... g24      svprec      12      7
2
9      1  6001 202500      14 060012025... g24      svprec      12      7
2
10     1  6001 202500A      14 060012025... g24      svprec      12      7
2
# i 51,113 more rows
# i 75 more variables: TOTREG <dbl>, DEMREG <dbl>, REPREG <dbl>, AIPREG <dbl>,
# GRNREG <dbl>, LIBREG <dbl>, NLPREG <dbl>, REFREG <dbl>, DCLREG <dbl>,
# MSCREG <dbl>, TOTVOTE <dbl>, DEMVOTE <dbl>, REPVOTE <dbl>, AIPVOTE <dbl>,
```

```
# GRNVOTE <dbl>, LIBVOTE <dbl>, NLPVOTE <dbl>, REFBOTE <dbl>, DCLVOTE <dbl>,  
# MSCVOTE <dbl>, PRCVOTE <dbl>, ABSVOTE <dbl>, ASSAIP01 <dbl>,  
# ASSDEM01 <dbl>, ASSDEM02 <dbl>, ASSREP01 <dbl>, ASSREP02 <dbl>, ...
```

It seems like based on the output that assembly election had many precincts featuring 2 candidates for at least 1 party. The same can be said about the senate and congress elections.